



Society of Petroleum Engineers

SPE-227574-MS

Remote Sensing and Machine Learning Framework for Natural Hydrogen Exploration

Taqi Alyousuf, Sustainable Geophysics Team, Geophysics Technology Division, EXPEC ARC, Saudi Aramco;
Hossain Sheikh, Exploration Geospatial & Data Management Solution, Saudi Aramco; Daniele Colombo,
Sustainable Geophysics Team, Geophysics Technology Division, EXPEC ARC, Saudi Aramco

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227574-MS](https://doi.org/10.2118/227574-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This study explores the application of remote sensing and machine learning techniques to detect surface indicators of hydrogen seepage in the Proterozoic Arabian Shield, with a focus on ophiolite formations. Utilizing Sentinel-2 satellite imagery from the European Space Agency and Digital Elevation Models (DEMs), we applied geospatial analysis and advanced deep learning algorithms to identify fairy circles and other circular surface anomalies potentially linked to subsurface hydrogen migration and surface seepage. The results demonstrate that machine learning can effectively detect these features across large areas, providing valuable insights into regions that may warrant further field investigation and validation. However, the study also highlights the need for higher-resolution imagery and high-resolution DEMs to improve detection accuracy, as well as the computational challenges of handling large datasets. The findings emphasize the potential of integrating remote sensing with machine learning to enhance the efficiency and effectiveness of hydrogen seepage exploration, offering a novel approach for large-scale geological surveys in regions with potential subsurface hydrogen resources.

Introduction

The exploration of natural hydrogen is in its infancy, hindered by limited understanding of its genesis, migration, and storage mechanisms. However, notable discoveries are catalyzing interest. For example, substantial hydrogen accumulations in Mali, attributed to interactions between water and iron-rich minerals in high-temperature geological settings, have demonstrated the feasibility of natural hydrogen production (Prinzhofer et al., 2018). Similarly, Australia has issued at least 18 exploration licenses as of early 2022, signaling increasing global interest in this resource (PV Magazine, 2022).

The significance of Mali's discovery is further highlighted by Renaud (2014) presenting a detailed study of the Timétrine Massif in northeastern Mali, which provides critical geological insights into hydrogen generation processes. The Timétrine region, located west of the Pan-African suture zone, exhibits a unique lithologic succession comprising Mesoproterozoic cratonic formations, Fe–Mg-rich schists, and ultramafic massifs of serpentized mantle peridotites. These ultramafic massifs, inserted as olistoliths

within sedimentary sequences, are the result of mantle exhumation during Neoproterozoic rifting and crustal thinning. The interaction of ultramafic rocks with water through serpentinization is a well-documented mechanism for molecular hydrogen generation. This process, coupled with the region's tectonic and sedimentary evolution, has created favorable conditions for natural hydrogen accumulation.

The discoveries in Mali are groundbreaking because they validate the concept of hydrogen-rich systems in ultramafic and rift-related settings, providing a tangible model for exploration efforts globally. By showcasing the geological and geochemical conditions necessary for hydrogen formation, the Mali example sets a benchmark for identifying hydrogen-prone regions. This discovery not only advances the scientific understanding of natural hydrogen systems but also positions Mali as a key player in the emerging hydrogen economy, offering a roadmap for exploration in similar geological settings worldwide.

Research has proposed several mechanisms for natural hydrogen generation. These include serpentinization in ultramafic rocks, radiolysis of water by radioactive decay in granitic systems, microbial activity, and cataclasis during faulting (Zgonnik, 2020; Saily et al., 2024). Additionally, deep primordial hydrogen from Earth's mantle, estimated at an annual flux of 23 Tg, underscores the vast but underexplored potential of this resource (Zgonnik, 2020).

Geologically, hydrogen occurrences are linked to three primary tectonic settings: extensional zones (e.g., mid-ocean ridges), compressional zones with ophiolitic nappes such as in Oman (Neal and Stanger, 1983), and stable intracratonic basins atop Precambrian basement rocks (Moretti, Brouilly, et al., 2021). Among these, extensional zones are less favorable for hydrogen accumulation due to pervasive faulting and fluid escape pathways (Wright, 1998). Identifying effective reservoirs and seals remains critical for commercial exploitation, mirroring challenges in petroleum system exploration (Dugamin et al., 2019). Despite growing interest, hydrogen exploration faces significant challenges, including identifying viable reservoirs and cost-effective exploration techniques. Hydrogen's chemical reactivity and mobility complicate long-term storage and detection. Zhang and Li (2024) highlight the emerging role of geophysics in exploring and monitoring natural and stimulated geologic hydrogen as a clean energy source, differentiating it from blue and green hydrogen. They emphasize a shift in hydrogen exploration toward source rock-focused approaches and propose innovative geophysical methods, including multi-physics integration, advanced data acquisition, and machine learning, to advance geologic hydrogen research and utilization over the coming decades.

The objective of this study is to develop a methodology for the exploration of natural hydrogen, focusing on identifying surface indicators of hydrogen. Specifically, the study aims to employ advanced multispectral remote sensing techniques, using datasets such as Sentinel-2, and ASTER, to detect and map potential hydrogen seepage zones through spectral signatures indicative of surface hydrogen release.

Remote Sensing to Detect Fairy Circles as Surface Indicators of H₂ Seepage

Remote sensing has become an essential tool for large-scale regional exploration in the search for potential hydrogen seepage areas. Satellite-based multispectral data provide an efficient means of gathering valuable information over extensive regions, allowing for the identification of surface anomalies that may be linked to subsurface hydrogen migration. Multispectral data are particularly beneficial for detecting variations in surface temperature, vegetation health, and land cover, all of which can serve as indirect indicators of hydrogen seepage from deeper reservoirs.

Fairy circles, circular barren patches surrounded by vegetation, have emerged as key surface indicators of hydrogen (H₂) seepage in diverse geological settings. These formations, hypothesized to result from subsurface gas escape, including H₂, alter soil properties, affecting vegetation growth and creating visible barren zones. Studies, such as those by Moretti, Brouilly, et al. (2021); Moretti, Prinzhofer, et al. (2021), suggest that these depressions are globally associated with H₂ seepage, with structures. Fairy circles are often associated with high hydrogen soil gas measurement and have been described in numerous countries such as Russia, Brazil, Mali, Australia, and the United States (Larin et al., 2015; Zgonnik et al., 2015;

Prinzhofer et al., 2019; Frery et al., 2021; Moretti, Brouilly, et al., 2021). Dimensions of fairy circles are generally distributed with mean diameters ranging from 150 to 200 meters, although variations, coalesced structures, or elliptical shapes are observed. Some depressions may feature raised rims, while others do not. Temporal monitoring, as illustrated in Figure 1, highlights significant variability in H₂ emission rates from fairy circles over time and depth.

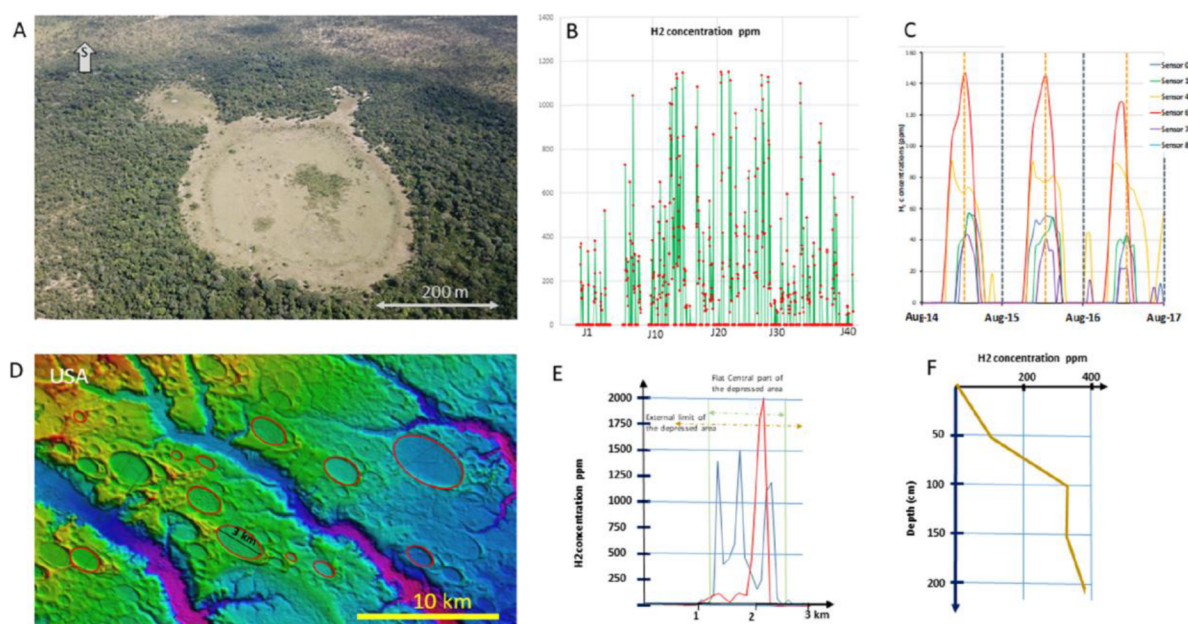


Figure 1—Synthesis of fairy circle emissions characteristics. It shows (a) fairy circle in Brazil illustrating the effect on surrounding vegetation, (b) temporal variability over a 5-week period from a sensor monitoring a structure in Brazil, (c) daily variations recorded by six sensors installed on the same structure in Brazil, showing a 24-hour period of measurement, (d) variability in size of emitting structures observed in the USA, (e) two profiles from Lake Podowoye, Russia, illustrating the spatial variability of hydrogen concentration values between the center and the rim of the structure, with measurements taken over two days, and (f) example of hydrogen concentration variation with depth in the USA (from Myagkiy et al., 2020)."

The analogy to hydrocarbon exploration provides additional insight into these features. Similar to how surface seepages of methane (CH₄) and oil were historically used to locate hydrocarbon reservoirs, H₂ seepages represented by fairy circles may signal underlying accumulations. Gas chimneys in seismic data, often associated with hydrocarbon seepage, provide a comparable model for understanding H₂ migration. These chimneys, active or inactive, may reach the surface and form pockmarks, similar to fairy circles. For H₂, such structures are associated with local emissions and linked to ultramafic rocks and fault zones, which facilitate hydrogen generation and migration. Frery et al. (2021) identified persistent H₂ seepage in Australia along the Darling Fault, underlining the geological correlation between fairy circles and ultramafic rocks, fault zones, and subsurface hydrogen systems. Additionally, Mosquera-Rivera et al. (2024) leveraged remote sensing, using Principal Component Analysis (PCA) and band ratio methods, to analyze fairy circles. Their findings showed correlations with ferric iron and water indices, suggesting that integrating high-resolution geophysical data with advanced classification algorithms could enhance the detection of hydrogen seepage. These studies demonstrate the value of fairy circles as low-cost, accessible surface indicators for early-stage hydrogen exploration.

The study of fairy circles and their association with surface hydrogen seepage provides important implications for deeper hydrogen exploration. These surface features offer a direct link between near-surface hydrogen migration and deeper subsurface hydrogen reservoirs. This predictive capability is essential for guiding further exploration efforts, particularly in regions where traditional exploration methods may be costly or difficult to implement.

Processing workflow

The workflow for hydrogen exploration through geospatial data analysis (Figure 2) integrates multiple data layers to identify potential hydrogen seepage zones and surface indicators such as fairy circles. The process initiates with Data Acquisition, where remote sensing data and digital elevation models (DEM) provide the foundational imagery and topographic context essential for regional hydrogen exploration. These initial datasets yield Data Derivatives, including multi-band images that enhance spectral information, slope attributes that highlight geological formations, and geology map classifications that segment the area based on known rock types and structures. Moving to Geospatial Analysis, the workflow combines these geological derivatives with advanced image classification methods and trained data, improving the detection of subtle anomalies that may indicate hydrogen seepage. This analytical stage is critical for isolating regions of interest where geochemical signatures or surface features could be linked to subsurface hydrogen migration. The final Results stage synthesizes these findings to map areas with high potential for hydrogen seepage and identifies unique surface features—fairy circles—that could suggest active or past hydrogen release. Detailed characterization of the identified structures can then follow with high resolution surveys (e.g., LiDAR). Geochemical analysis of the near surface rocks through shallow drilling and measurements and monitoring of H₂ emissions with hydrogen gas detectors can prove the existence of H₂ emissions and quantify the concentrations. This integrated geospatial approach provides a low-cost, effective means of hydrogen reconnaissance, enabling a targeted focus on areas with promising geological conditions.

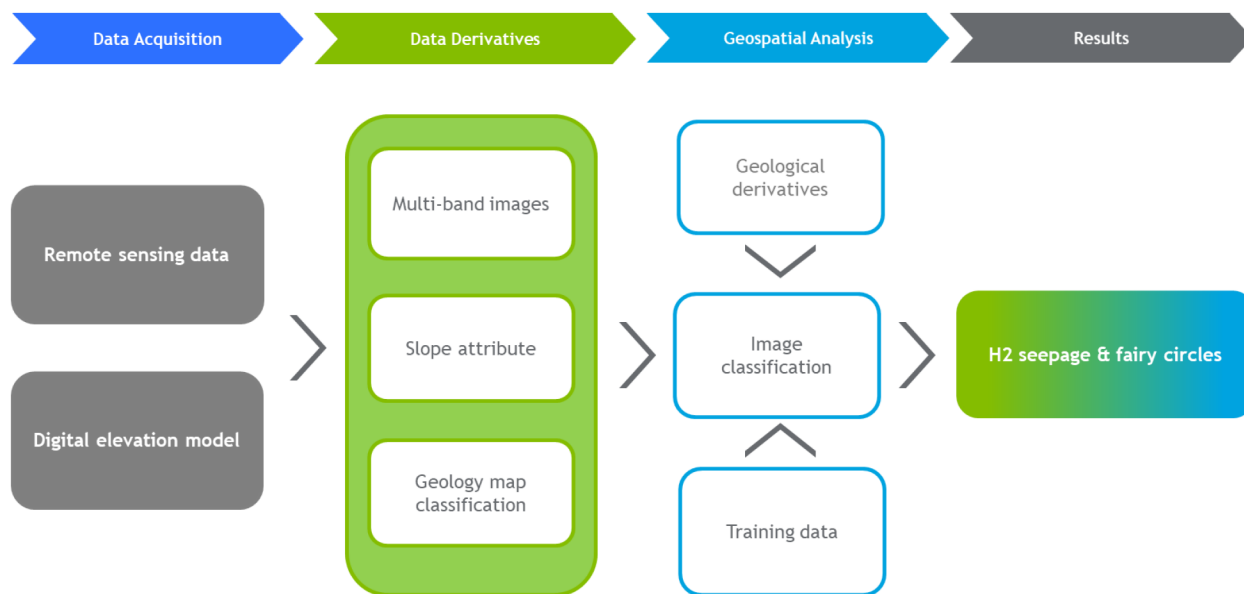


Figure 2—Workflow for hydrogen exploration using geospatial data analysis. The process begins with Data Acquisition, which includes remote sensing and digital elevation model (DEM) data collection. From these, Data Derivatives are generated, such as multi-band images, slope attributes, and geology map classifications that reveal key geological features. Geospatial Analysis then integrates these geological derivatives with image classification techniques and trained datasets to identify patterns. The results of this process highlight potential hydrogen seepage zones and identify surface indicators, such as fairy circles, which are linked to subsurface hydrogen migration.

By integrating these preprocessing and fusion techniques, we were able to enhance the sensitivity and accuracy of remote sensing analyses, facilitating the identification of potential hydrogen seepage areas over large geographical areas. This combination of sophisticated data correction and fusion methods is essential for the effective use of remote sensing in hydrogen exploration, particularly in regions with complex geological and environmental conditions.

Experimental Evidence: Results of Remote Sensing Analysis

This section presents our analysis of potential hydrogen seepage in the Proterozoic Arabian Shield using remote sensing techniques applied to five distinct areas, with a focus on ophiolites. We integrated multispectral remote sensing data from Sentinel-2, and ASTER satellites to identify surface features indicative of hydrogen release in northwest Saudi Arabia. Data fusion techniques were employed to enhance spectral resolution and isolate key surface signatures associated with hydrogen migration.

In this study, remote sensing data acquisition over the Arabian Shield was carried out using Sentinel-2 satellite imagery from the European Space Agency to capture multispectral data across the area of interest (Figure 3). A Digital Elevation Model (DEM) was also acquired to generate terrain derivatives, with slope analysis and geomorphological filtering applied to exclude regions with extreme slopes and extensive sand dune coverage. This preprocessing ensured that the analysis was focused on geologically relevant zones that may exhibit surface features indicative of hydrogen seepage. Building on this, four pilot studies were conducted at locations coincident with known ophiolite formations to investigate the morphological characteristics of these areas and test the application of deep learning techniques in identifying surface features related to hydrogen seepage. These pilot studies aimed to better understand the geological and morphological context of hydrogen migration in these regions and evaluate the effectiveness of advanced machine learning methods in detecting potential seepage zones.

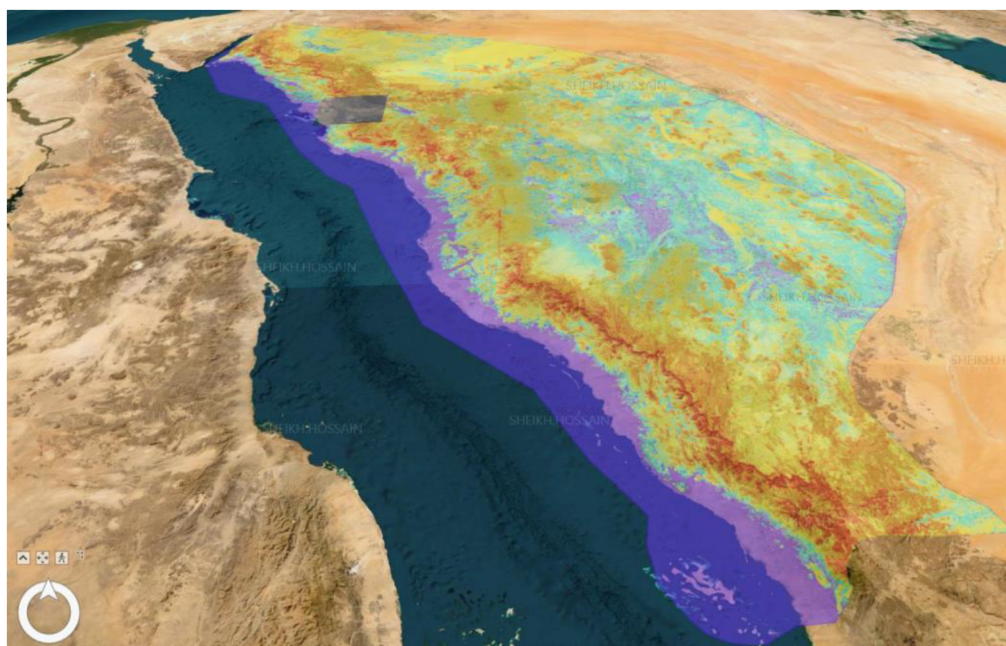


Figure 3—Acquisition and preprocessing of remote sensing data over the Arabian Shield. Sentinel-2 satellite imagery from the European Space Agency was used to capture multispectral data over the area of interest. A Digital Elevation Model (DEM) was also acquired to generate terrain derivatives. Slope analysis and geomorphological filtering were applied to exclude areas with extreme slopes and extensive sand dune coverage, ensuring focus on geologically relevant zones for hydrogen seepage investigation.

Figure 4 illustrates the process of training images for a machine learning (ML) approach aimed at identifying fairy circles. The deep learning algorithm was utilized, employing manual training to teach the model to recognize patterns associated with these surface features. The trained algorithm successfully identified fairy circles, demonstrating the potential of machine learning techniques for detecting morphological anomalies that could be indicative of subsurface hydrogen seepage. This method highlights the power of deep learning in automating and enhancing the detection of geological features in remote sensing imagery.

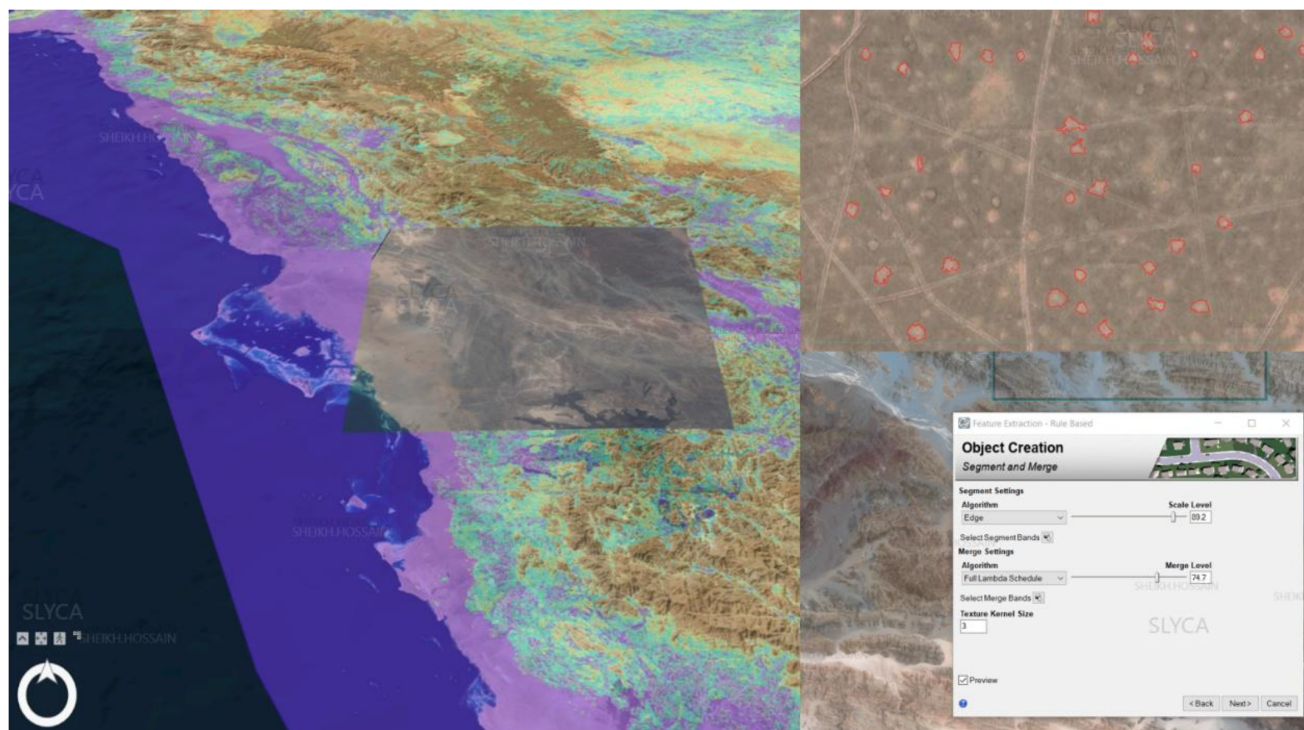


Figure 4—Images trained for machine learning (ML) approach to identify fairy circles. The ENVI deep learning algorithm was employed with manual training to detect surface features, resulting in the identification of fairy circles through the ML model.

In [Figure 5](#), the results of the automated detection of fairy circles are presented, with these surface anomalies identified through a combination of geospatial analysis and remote sensing techniques. Using the deep learning algorithm trained manually, circular patterns on the surface were detected, which may indicate potential hydrogen seepage zones. The map illustrates clusters of these anomalies within the study area, highlighting key regions for prioritization in further field validation and investigation. This automated detection process not only enhances the efficiency of surveying vast areas but also supports the strategic targeting of sites that warrant more detailed exploration.

The presence of fairy circle in the northern Arabian Shield, as also suggested by [Guirado et al. \(2023\)](#) using global systematic survey, provides compelling evidence supporting our site selection for exploration. Guirado's research identifies 263 sites across 15 countries and three continents where these distinctive patterns occur, including regions with arid climates, high precipitation seasonality, and sandy, nutrient-poor soils. These findings align with the environmental characteristics of the northern Arabian Shield, further validating its potential as a significant target area.

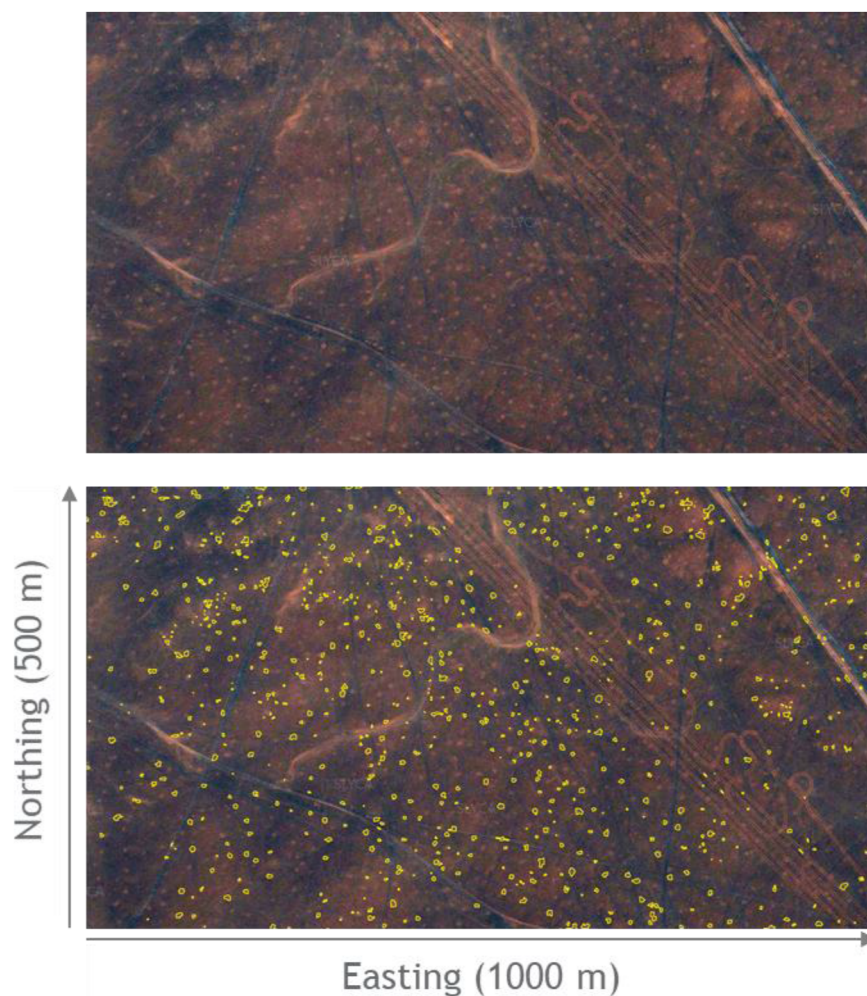


Figure 5—Map highlighting the identified areas with fairy circles, automatically detected using geospatial analysis and remote sensing techniques. Deep learning algorithm was used with manual training. The algorithm identifies circular surface anomalies that are potential indicators of hydrogen seepage, shown here as clusters within the study area. This automated detection enhances efficiency in surveying vast regions and helps prioritize sites for further investigation and field validation.

Conclusion

This study demonstrates the potential of remote sensing and machine learning techniques for detecting surface indicators of hydrogen seepage in the Proterozoic Arabian Shield. Through the integration of Sentinel-2 satellite imagery, Digital Elevation Models (DEMs), and advanced deep learning algorithms, we successfully identified fairy circles and other circular surface anomalies that may be indicative of subsurface hydrogen migration. The use of geospatial analysis and ML approaches allowed for the efficient detection of these features across large areas, highlighting key regions for further investigation and field validation. While the results show promise, the study also identifies several areas for improvement, including the need for higher-resolution imagery and high-resolution DEMs to capture finer details and enhance feature detection. Additionally, computational challenges related to processing large datasets suggest the need for more advanced resources to optimize machine learning workflows. Overall, this research underscores the value of combining remote sensing and machine learning for large-scale geological surveys, providing a powerful tool for identifying and prioritizing areas of interest in hydrogen seepage exploration. Future work will focus on refining these techniques and addressing the identified challenges to improve the accuracy and efficiency of surface anomaly detection.

References

- Dugamin, E., L. Truche, and F. V. Donze. 2019, Natural hydrogen exploration guide. *ISRN Geonum-NST*, **1**, 16.
- Frery, E., L. Langhi, M. Maison, and I. Moretti. 2021, Natural hydrogen seeps identified in the north perth basin, western Australia. *International Journal of Hydrogen Energy*, **46**, no. 61, 31158–31173.
- Guirado, E., M. Delgado-Baquerizo, B. M. Benito, J. L. Molina-Pardo, M. Berdugo, J. Martínez-Valderrama, and F. T. Maestre. 2023, The global biogeography and environmental drivers of fairy circles. *Proceedings of the National Academy of Sciences*, **120**, no. 40, e2304032120.
- Larin, N., V. Zgonnik, S. Rodina, E. Deville, A. Prinzhofer, and V. N. Larin. 2015, Natural molecular hydrogen seepage associated with surficial, rounded depressions on the European craton in Russia. *Natural Resources Research*, **24**, 369–383.
- Moretti, I., E. Brouilly, K. Loiseau, A. Prinzhofer, and E. Deville. 2021, Hydrogen emanations in intracratonic areas: new guide lines for early exploration basin screening. *Geosciences*, **11**, no. 3, 145.
- Moretti, I., A. Prinzhofer, J. Françolin, C. Pacheco, M. Rosanne, F. Rupin, and J. Mertens. 2021, Long-term monitoring of natural hydrogen superficial emissions in a brazilian cratonic environment. *Sporadic large pulses versus daily periodic emissions. International Journal of Hydrogen Energy*, **46**, no. 5, 3615–3628.
- Mosquera-Rivera, J. E., J. M. Jiménez-Vergara, C. Alberto, B. Philip, and H. Morales. 2024, Preliminary Remote Spatial Analysis of Fairy Circles: an Approximation of Hypersectral and Geophysical Data from Hydrogen Seeps. *First Break*, **42**, no. 6, 65–78.
- Neal, C., and G. Stanger. 1983, Hydrogen generation from mantle source rocks in Oman. *Earth and Planetary Science Letters*, **66**, 315–320.
- Prinzhofer, A., C. S. T. Cissé, and A. B. Diallo. 2018, Discovery of a large accumulation of natural hydrogen in Bourakebougou (Mali). *International Journal of Hydrogen Energy*, **43**, no. 42, 19315–19326.
- Prinzhofer, A., I. Moretti, J. Françolin, C. Pacheco, A. d'Agostino, J. Werly, and F. Rupin. 2019, Natural hydrogen continuous emission from sedimentary basins: The example of a Brazilian H₂-emitting structure. *International Journal of Hydrogen Energy*, **44**, no. 12, 5676–5685.
- Renaud, C. 2014, Nature and evolution of Neoproterozoic ocean-continent transition: Evidence from the passive margin of the West African craton in NE Mali. *Journal of African Earth Sciences*, **91**, 1–11.
- Saidy, K., M. Fawad, S. A. Whattam, A. A. Al-Shuhail, A. A. Al-Shuhail, M. Campos, and F. A. Sulistyohariyanto. 2024, Unlocking the H₂ potential in Saudi Arabia: Exploring serpentinites as a source of H₂ production. *International Journal of Hydrogen Energy*, **89**, 1482–1491.
- Zgonnik, V. 2020, The occurrence and geoscience of natural hydrogen: A comprehensive review. *Earth-Science Reviews*, **203**, 103140.
- Zgonnik, V., V. Beaumont, E. Deville, N. Larin, D. Pillot, and K. M. Farrell. 2015, Evidence for natural molecular hydrogen seepage associated with Carolina bays (surficial, ovoid depressions on the Atlantic Coastal Plain, Province of the USA). *Progress in Earth and Planetary Science*, **2**, 1–15.
- Zhang, M., and Y. Li. 2024, The role of geophysics in geologic hydrogen resources. *Journal of Geophysics and Engineering*, gxae056.